

Saumil Agarwal

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Summary

Member of Technical Staff with 4 years of experience building and owning production systems across distributed infrastructure and network security. Proven track record of leading cross-functional initiatives, designing scalable systems, and delivering customer-facing features end-to-end. Seeking to bring this systems and engineering background to startups building impactful, production-grade applied AI products.

Experience

NUTANIX, INC.

Bengaluru, India

Member of Technical Staff - 2

Aug 2024 - Present

- Implemented **RDMA QoS** on **NVIDIA Mellanox** ConnectX NICs using PFC and ECN across SR-IOV network interfaces. Enabled lossless data transfer for **AI workloads** and **large scale storage replication**, while prioritizing latency-sensitive traffic over background transfers.
- Built and deployed **Jira to GitHub on Autopilot**, an **end-to-end agentic AI workflow** that automated bug reproduction, log analysis, code fix generation, testing, and code review to generate production-ready pull requests. Enabled parallel bug processing, reduced repetitive manual effort, and *saved 2 to 3 hours* of engineering effort per issue.
- Designed and established the initial framework for a **Go microservice for security policy management from scratch**, replacing a Python based architecture to reduce memory consumption, improve execution speed, and enable greater parallelism at scale. Reduced its build pipeline from *2 hours to 30 minutes, a 75% reduction*.
- Led a *four person* team to design & implement **tenant isolation across our network security platform**, preventing cross-tenant data access across all product components. Coordinated integration across **12 engineering teams** and delivered the feature within a short release timeline.
- Reworked a **distributed network telemetry pipeline** using **NATS JetStream**, improving data synchronization and building a reusable collector service that enabled multiple engineering teams to efficiently consume telemetry concurrently.
- Delivered **FQDN based rules** and **IP address exclusions**, addressing a long standing customer request, reducing policy management overhead, and closing a key competitor gap. Owned the features end-to-end across design, API, implementation, production quality, and on-call support.
- Owned **production releases end-to-end**, including release packaging and offline deployments. Led on-call triage for production blocking incidents, including escalations from *regulated banking and government customers*, diagnosing critical defects under time pressure, coordinating fixes across teams, and driving incidents through resolution while minimizing customer impact and maintaining customer confidence.

Member of Technical Staff - 1

Jul 2022 - Aug 2024

- Identified and resolved long standing out-of-memory failures in a distributed network telemetry collector, **tracing memory leaks** and task contention across data collection and monitoring components. Improved reliability and operability for large scale customer deployments.
- Diagnosed a release blocking **issue introduced by an operating system kernel change**, tracing the failure beyond the application layer into a large, externally maintained kernel codebase. Identified the root cause & drove the issue to resolution, preventing significant customer impact.
- Led a team to a **company wide hackathon win** by identifying a key gap in the customer security workflow and delivering an end-to-end solution within three days. Built dashboard analytics and additional Layer 7 security capabilities using insights from customer interactions, while evaluating implementation cost, adoption potential, and commercial viability.

Member of Technical Staff - Intern

Jan 2022 - Jun 2022

- Automated the product release and deployment pipeline, cutting the shipment time from *several weeks to a few days*.
- Designed migration from Apache Kafka to an IPFIX export pipeline, reducing resource usage and improving scalability for large deployments.

WELLS FARGO BANK

Remote

Software Engineer Intern

May 2021 - Jul 2021

- Developed a cloud application for loan default risk assessment, enabling underwriters to evaluate borrower risk through model predictions.
- Built an **explainable machine learning** pipeline, training interpretable models that surfaced the key factors behind loan approval decisions.

DALHOUSIE UNIVERSITY, NOVA SCOTIA, CANADA

Remote

Machine Learning Research Assistant

Jul 2021 - Oct 2021

- Modeled human aging using stochastic processes & machine learning on complex networks to predict aging categories from biological data.
- Awarded the **MITACS Globalink Scholarship** for the research project at Dalhousie University.

TWOFOLD EDITIONS - 3D PRINTING

India

Founder

July 2026 - Present

- Designed and manufactured 3D-printed collectibles, while managing product development, e-commerce, fulfillment, & customer experience.
- Website: www.twofoldditions.in

Education

Birla Institute of Technology and Science (BITS), Pilani

Pilani, India

Bachelor of Engineering (Hons.) in Computer Science and Minor in Data Science

2018 - 2022

9.09 CGPA (Received **Merit Scholarship**, for 2 consecutive semesters for being within top 3% of students)

Position of Responsibility - Core Member of **Srishtaara Makerspace** (Among the first few Library Makerspaces in India)

Skills

Languages & Systems • Go • Python • C/C++ • Distributed Systems • Network Security • Microservices • Linux

AI & Infrastructure • Agentic AI • Machine Learning • API Design • Performance Optimisation • Kubernetes • Monitoring & Visualisation